

Evaluator's Assessment Report

Problem Statement 26023 — AI-Powered Geological, Mining and other Reporting Solution for CMPDI/CIL
Team: Cod-Ons | Solution: Lanzey | Reviewed against: lanzey_solution.pptx

1. What This Report Is

This is written from the seat of an internal-hackathon evaluator (SPOC / faculty jury), not as a rewrite of your deck. It uses the criteria that actually decide who gets nominated out of an internal round — problem understanding, solution clarity, innovation, technical feasibility, a working prototype, impact, and presentation — and applies each one to Lanzey as it stands in your current PPT and in the workflow you described.

Read section 4 first if you're short on time — it's the criterion-by-criterion verdict with fixes.

2. The Evaluation Framework Being Used

These are the questions a jury is trained to ask at the internal round, and again (more strictly) at national pre-screening:

Criterion	What evaluators are actually checking
Problem understanding	Have you correctly read PS 26023 and named the actual CMPDI/CIL pain points — not a generic 'AI reporting tool' pitch?
Solution clarity	Can the team explain, in under two minutes and without the slide, what Lanzey does end to end?
Innovation / uniqueness	What does Lanzey do that a generic RAG-over-documents chatbot does not?
Technical feasibility	Can the stack named actually be built and demoed in the time available — internal round first, then 36-hour grand finale?
Working prototype	Is there something running, even a thin slice — not only slides?
Impact & relevance	Does the benefit map back to a metric CMPDI/CIL would actually track (reporting time, error rate, audit turnaround)?
Presentation & communication	Is the story easy to follow live, not just readable as a static PDF?

3. Your Workflow, As It Currently Reads

Reconstructed from slide 3 ("Lanzey Workflow") and your description, so you can check it's exactly what you mean to present:

Step	What happens
1. Login & Access	Role-based login for CIL, CMPDI and college/institution users — separate dashboards for geological admin, machinery supervisor, reserve/audit officer, etc.
2. Upload Documents	Field staff / officers upload reports, images, PDFs, scanned forms, sensor data.
3. AI Processing	OCR + NLP extracts text and structures the uploaded documents into tokens / structured fields.
4. AI Analysis & Prediction	Structured data is run through the geological, machinery-risk and environmental models to generate a draft assessment.
5. Officer Verification	A human officer reviews the AI output against the uploaded evidence (snapshots/images) and validates, corrects, or rejects it.
6. Verified Report	An auto-generated report is produced with the AI insight plus the human-verified evidence trail.
7. Send to CIL / CMPDI	The CMPDI/CIL officer does a second-level review and marks the report Approved / Suggest Changes / Return for Revision.

If this doesn't match what you intended to say, fix the sequence before the pitch — an evaluator will follow the arrows on the slide, not your verbal explanation, and any mismatch between the two is the first thing a sharp juror will catch.

4. Criterion-by-Criterion Verdict

Criterion	Where you're strong	What a jury will probe	Fix before you present
Problem understanding	Problem slide names the real CMPDI/CIL pain: 100–200 page reports, data scattered across PDF/Excel/DB, years of history, slow manual reporting.	PS 26023 is broad (geological + mining + "other" reporting). You haven't stated which modules you'll actually build for internal round vs. later.	Add one line: "For this round we are demonstrating modules X and Y; the remaining N modules follow the roadmap."
Innovation / uniqueness	"What makes Lanzey unique" panel (cross-document intelligence, domain-specific AI, source-grounded answers, historical trend intelligence) is a genuinely good differentiator list.	As written it still reads as "chatbot over documents." A jury will ask what stops this from being ChatGPT + file upload.	Lead with the mandatory human-verification + audit trail + risk-prediction layer — that's the part a generic RAG tool doesn't have. Say it first, not last.
Technical feasibility	Tech stack is concrete and reasonable (React/Tailwind, FastAPI, Postgres/PostGIS/TimescaleDB,	Full stack + 27 modules + geospatial + OCR + LLM + real-time dashboards is a large build. A jury will	State an MVP slice explicitly: e.g. "upload → OCR/token extraction → officer verification → one prediction module"

Criterion	Where you're strong	What a jury will probe	Fix before you present
	OCR+LLM, QGIS/Folium, Docker).	ask what's realistically demoable now.	as the thing you can actually show working.
Working prototype	Workflow and UI mockups exist and are detailed enough to build from.	Slides don't say whether anything is running yet. Internal-round juries eliminate mostly on this.	Bring whatever runs — even a fake-backend upload screen that returns a canned extraction — and show it live before the slides.
Human-in-the-loop / trust	Officer verification with snapshot evidence, and a second CMPDI-level approve/reject step, is a strong trust story for a PSU deployment.	The order of AI vs. human steps is not fully consistent between your slide (AI analysis, then officer verification) and how you described it verbally (officer verifies first, report sent, then AI predicts, then CMPDI decides).	Fix the sequence on paper first — pick one order and use it everywhere (slide, doc, pitch): who sees the AI output before whom, and at what point evidence is attached.
Impact & relevance	Before/after bar chart (reporting time, manual effort, risk detection, decision support) is exactly the kind of quantified impact juries want.	The numbers on that chart aren't sourced — a jury may ask where they came from.	Label the chart "illustrative / target improvement" unless you have a real baseline, or replace with a smaller claim you can defend.
Data readiness	Research slide cites real public sources (NGDR, Bhukosh, USGS, ERA5, MSHA, Sentinel-2, PARIVESH/MSS).	None of these give you actual CIL/CMPDI internal report data — your real workload is synthetic/sample data until a PSU MoU exists.	Say explicitly how you'll demo without real CMPDI data — synthetic reports styled like the real ones is a fine, honest answer.

5. Questions This Jury Would Likely Ask

- Which of the 27 modules in your full design are you actually building for this round?
- Walk me through one real document, end to end, right now — not the slide, the tool.
- At what exact step does a human see the AI's output before CMPDI does? Say the sequence out loud.
- Where does the "risk prediction" number come from — a trained model, or a rule you wrote?
- What happens if the officer disagrees with the AI? Show me that path, not just the happy path.
- Your before/after chart claims faster reporting and better risk detection — what are those numbers based on?
- CIL/CMPDI is a PSU handling sensitive data — where does the data live, and who can see it?

6. Before You Present — Checklist

- Pick and state your MVP scope for this round (don't imply all 27 modules are built).
- Lock one workflow sequence and make the slide, the doc, and your spoken pitch agree.
- Have something on screen that actually runs — even a stubbed upload-to-extraction flow.
- Be ready to defend or relabel every number on the impact chart.
- Prepare a one-line answer for "why not just ChatGPT with file upload" — lead with human verification + audit trail + domain risk prediction.
- Have a one-line data-sourcing answer: which parts use real public data (NGDR, Bhukosh, USGS, ERA5, MSHA) vs. synthetic CMPDI-style samples.

Prepared as a practice evaluation ahead of SIH 2026 internal-round judging — not an official SIH document.